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Department of Physics

What Quantum Trace Can a Heavy Particle Leave?

Curvature–Spectator Entanglement
and the Cosmological Collider

Step 7: Spectator Bispectrum

Group-B

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1 Mixed contractions and the spectator three-point function

We use $H_I = \lambda_3 \int \mathrm{d}^3x\, a^4\sigma^3/3!$, with $\lambda_3/H = 0.01$. The cubic interaction is treated to first order about the stable quadratic state. Define

$$K_k(\eta_f, \eta) = \sum_{a=1}^{2} q_k^a(\eta_f)\sigma_k^{a*}(\eta). \tag{1}$$

Wick's six contractions cancel 3!. If $\langle OH_I\rangle$ contains the product $K_1K_2K_3$, reversing the operator order gives its complex conjugate. Consequently, $-i(Z - Z^*) = 2\,\mathrm{Im}\,Z$, and

$$B_q(k_1, k_2, k_3) = 2\lambda_3 \int \mathrm{d}\eta\, a^4\,\mathrm{Im}(K_{k_1}K_{k_2}K_{k_3}). \tag{2}$$

For $(k_1, k_2, k_3) = (r, 1, 1)$ and $z = -k_S\eta$, all legs share the final physical time. The short argument is z_f and the long argument is rz_f. A final long-leg value at z_f would evaluate a different physical time. Let $C_S(z) = \sum Q_a(z_f)\Sigma_a^*(z)$ and $C_L(z) = \sum Q_a(rz_f)\Sigma_a^*(rz)$. With $H = k_S = 1$,

$$F(r) = \frac{HB_q}{P_q(r)P_q(1)} = \frac{2\lambda_3}{p_L p_S}\int_{z_f}^{\infty}\frac{\mathrm{d}z}{z^4}\,\mathrm{Im}[C_L(z)C_S^2(z)], \tag{3}$$

where $p_L = \sum |Q_a(rz_f)|^2$ and $p_S = \sum |Q_a(z_f)|^2$. The r^{-3} factors cancel. Power spectra alone cannot replace these complex contractions.

2 Regulated pilot and control tests

The pilot uses $M = 2$, $g = 0.1$, $z_f = 0.01$, $X = 800$ and the sixth-order ultraviolet initial state. The displayed cumulative integral includes $e^{-0.025(2+r)z}$ and stops at $z_{\max} = 320$. These finite-regulator values are inputs to Step 8, rather than the final extrapolation.

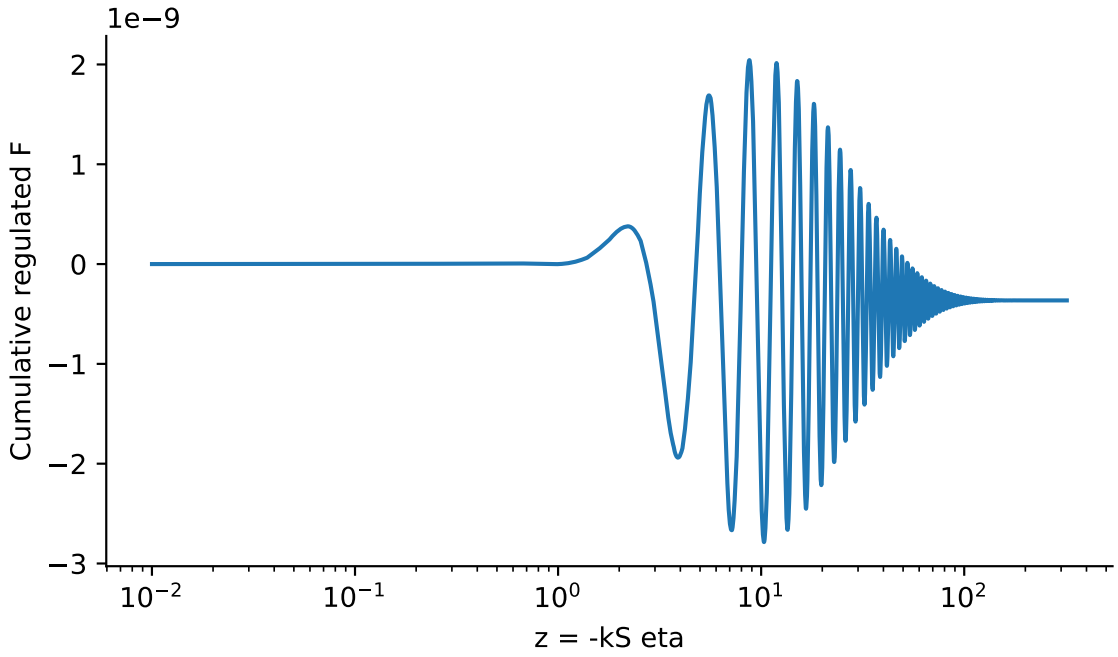


Figure 1: Cumulative regulated bispectrum for $r = 0.03$. Keeping its time dependence makes cancellations and an unfinished early-time remainder visible.

The calculation returns exactly zero for $g = 0$ and for $\lambda_3 = 0$. Doubling λ_3 doubles the result. The weak-mixing table compares $g = 0.025, 0.05, 0.1$ through F/g^3, testing the expected leading cubic mixing scaling. The saved complex mode file contains the rows $(Q, \Pi_q, \Sigma, \Pi_\sigma)$, two mode columns and an oscillation-resolving early-time grid. The integrand archive stores complex integrand values, quadrature nodes and weights, final powers and settings. These are retained because a single final number cannot independently expose a phase or cancellation error.

References

[1] X. Chen and Y. Wang, *Quasi-Single Field Inflation and Non-Gaussianities*, JCAP 04 (2010) 027, arXiv:0911.3380.

[2] N. Arkani-Hamed and J. Maldacena, *Cosmological Collider Physics*, arXiv:1503.08043.

Code and supporting data: step7 on GitHub.

Contributions (equal)

Member	Contribution
Priyansh Rawat	Mixed-contraction and bispectrum derivation.
Samarth Bhatnagar	Common-final-time and zero/cubic controls.
Mayank Sinha	Complex-mode integration, saved phases and figures.